

"Physics as Information Processing" ~ Ander Aguirre ~ Discussion 3

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Hello and welcome. It's July 29th, 2023 and we're here in session three discussion of the course Physics's Information Processing at the Active Inference Institute. We're having a follow-up discussion on Chris Fields' third lecture on Quantum Reference Frames and we're looking forward to your participation. So if you're watching along live and you've been invited to the calendar event, it's not too late to jump in. If you're watching on YouTube, you can write any questions or comments in the live chat with Dean and Ender here. We're going to be looking at some of the key points and taking it in a few different directions, sharing our reference frames, making a shared one. So, Ender would be awesome to have you zoom in a little bit on the slide and then feel free to take us through some of the key points so that we can see a little bit how you're seeing it. Thank you. That looks good? Yep. Okay. So let's start by reviewing the last lecture from a couple weeks ago and then we can open up the discussions if there's any questions that appear in the live chat. We can address those. All right. So in lecture two, Chris introduced this idea of the information processing between system A and B, you know, where they interact via qubits on a boundary and they have reference frames that give meaning to this communication. You want to think of information processing as a communication channel, the way Shannon thought about it. So that was in lecture two. And lecture three, a couple weeks ago, tried to make this more explicit by emphasizing the idea that this has to be physically embodied somewhere. Okay. And I think that because we are so used to information being fungible, in no small part because of the internet and everything, it's a little hard to understand that first when the claim is made that you have a QRF that it's non fungible, or you have a piece of quantum information that's non fungible or things like the no cloning theorem. We're so used to information being fungible to it being zeros and ones that it's a bit of a change in mindset at first, but that's exactly what lecture three introduced. This was all, all stuff, right?

Could you maybe just unpack that a little bit? What would it mean for information to be fungible or what's that stance like? And then what does it mean for information to be non fungible? And what is that stance like? Yeah, I'm out to get there. So let me let me scroll down the slides. Because that's addressed here. So I was maybe I should have kept this in the beginning. But anyways, session two talked about how information theory makes this simple and obvious. Okay. And in session three, we talk about reference frames, and mechanistically how all of this works. So yeah, let's address that question of the difference between between, hold on one second, let me use my mouse to scroll down here. Okay, so that's, that's a little faster, right? Okay, so can you restate your question, Daniel? Yeah, what would it mean for information to be fungible or non fungible? People might be familiar with that in the economic setting, with non fungible tokens and all of that. But what are we talking about with informational fungibility? I think it basically comes down to whether you can replicate, I mean, it's basically the difference between classical and quantum information, as far as I know. Fungible information is everything that you

can basically turn into a message of zeros and ones. It's, you know, how you transmit it may be different, you know, the channel, the physical hardware, and so on. But, but at the end of the day, it's something that you can write on a boundary, and something that you can transmit as zeros and ones. Whereas quantum information, you literally have to send the hardware, the piece of hardware, the QRF. At least that's how I understand it. So, a simple answer might be that it's information encoded in a piece of software versus a piece of hardware, but I don't, I don't think that analogy is perfect.

But what do you, yeah, Dean, what do you think?

Dean Mitchell- Well, here's what I'm curious about. And again, you guys can help me with this.

Dean Mitchell- As opposed to reference frames, I oftentimes default to frame of reference. And I know that in one of the live streams, what I tried to bring out was the idea that there was an ice cream cone looking at an ice cream truck. And the question was always whether or not which, which one of those two entities is that one is in the foreground, and one is in the background. And in this slide that Anderhazza, the question is, is the non-fungibility because only A and B can decide whether the A, the VA as seen by B, the A is deeper into that image, based on our frame of reference, and the B is closer, or is the B the one that's actually deeper, and the A is the one that's in the foreground. And I was wondering whether the idea of non-copyability is because of that depth of field aspect that cannot be determined, unless it's literally handed over the way that Chris was describing it.

Dean Mitchell- I think that's basically accurate. I think, I'm not sure I can answer the question, but I think, I have the feeling that Chris, well Chris said this in the lecture that, I mean, at the end of the day, the QRF is somewhere in your nervous system. And here, whether, I mean, you bring up a good point, right? It looks like this slide is a bit of an optical illusion. Whether you see A or B deeper, I guess it depends on something going on inside your head. If you, this picture is so symmetrical that if you focus enough on it, you can change your opinion.

Dean Mitchell- Well, just to even connect that to what we're talking about, the classical fungible information on the holographic screen, on the blanket, and then the quantum cognitive rotation, is those two images are on the screen identical. And then whether it gets a quantum rotation such that the three points of the axes are closer to us than the intersection on that kind of tetrahedron, or the other way around, whether the ballerina is rotating one way or the other, that's not on the boundary. So the information about the depth of field is part of the QRF or part of the enactment of the QRF, and it's not written on the boundary.

Dean Mitchell- And that goes back to the preparation question and not just the measuring.

Dean Mitchell- Yeah. Like in the preparation, if your whole enculturation or the last five minutes, it was entrained or you were holding an object that predisposed you to think that the vertex of the intersection was further away, that's the preparation that sets your prior so that when you get this ambiguous artifact, you rotate it into the most natural position cognitively. And that's the preparation that sets you up for measurement. No measurement can happen without a preparation. What do you think, Andrew? Andrew

Andrew Pérez- Sure. Yeah. Everything you said makes sense. So perhaps let me let me recall what was said around the slides. There was a, in this whole spirit or philosophy of making, emphasizing that information is physical, you know, from qubit, this whole thing.

Andrew Pérez- One of the glaring omissions in physics was the embodiment of the reference frame, which was treated as an abstraction, you know, in classical physics, such as by Galileo. But maybe even Einstein, even if he, my hunch is that he did have more of this instinct of, you know, particularly in all those thought experiments of trying to chase the light ray, you are chasing the light ray, right? So at the end of the day. But I mean, this is a whole process of making the observer and the reference frame more and more explicit and more and more physical. Because at the end of the day, it has to, right? If you start with this assumption, that solipsistic assumption, almost that, you know, all you see and, you know, it's part of the same material ether that you are, then, you know, then you are part of the universe that you're observing something that is out there.

Andrew Pérez- So you need to make, include a theory of what it means for you to be observing that, right? And at the end of the day, the observer has to be physical. That's what was emphasized by Chris at the beginning of this lecture. So this jump from treating the reference frame as an abstraction to a real physical entity. And yeah, these are, these people have

Andrew Pérez- talked a lot about in these papers that are quoted there. But I should also remark, Andrew Pérez- if folks want to see this, it's coming up in high energy theory also in recent work of, in recent work of Sorkin and others, where they talk about quantum reference frames, very explicitly. So I don't, I don't have the citations off the bat, but it's definitely becoming an important thing in the last 20 years. This attempt to really explicitly treat the reference frame and the observer as a quantum entity, as a physical entity. Okay, so this, that's what I just said. Okay, the jump from treating things, reference frames, as abstractions to the physical objects.

Okay.

Andrew Pérez- And because we cannot make a perfect copy of quantum information, we say that these are unique and clonable and unclonable. So in order to recreate an experiment perfectly, this is the, this is addressed as a part of fungibility or not, right? Can Alice, if you have some sort of quantum experiment, and Alice and Bob are, you know, it's the usual EPR, thought experiment or whatever. Alice and Bob are separated by light years. Is it enough if they want to recreate the same, exact, the same experiment for Alice to send a string of zeros and ones? Well, if you had an experiment where that's enough, then you can recreate it with fungible information. If not, then you need to somehow send non-fungible information, which really implies sending a piece of hardware.

Andrew Pérez- Can I, can I add another, another piece here? Look on the previous slide. So this looks really technical and scientific, but let's think about Alice's laboratory is Alice's brain, mind, and Bob's analogously. Alice is thinking about what to say, and then says it. And that information might be written in a chat form, it might be said verbally, it might be transmitted through digital or analog signal. But at some point,

it's going

through this classical informational interface, and then Bob's laboratory is like perceiving and unpacking it. And the question of semantic alignment is such that if Alice gives a shopping list of 10 items, if there's semantic alignment, Bob would be able to go to the store and then find those 10 items. And it made me think of the dictionary definition of left and right. So this gets back to our discussion of whether the vertex on these axes is coming towards us or away from us. So I looked up the dictionary definition of left. On, towards, or relating to the side of a human body or of a thing that is to the west when the person or that thing is facing north. So that's, I guess, one attempt to anchor what it means to be left. The second definition is the left-hand part, side, or direction. And you could imagine there's an analogous definition for right. The point is, you can't describe verbally leftness and rightness or chirality in general because its reference is an embodied perspective. And so if Alice said, okay, you're going to go down the street and then turn left at the stop sign, stop sign. How would Alice communicate to Bob what leftness meant at that stop sign versus rightness? And that's this question of flipping and the way that the quantum formalism accommodates that is through the wick rotation and the quantum cognitive events where the fungible, transmittable, holographic information gets rotated into the imaginary axis with the formalisms that enable it to be rotated at all. But that means you could rotate it left quote or rotate it right.

Yeah, I think that's, that's a very good explanation. I don't know if I have much to add. Dean, you wanted to say something? Well, so, but I can't, I don't know the answer to this, but to take what Daniel just said, to take the basic concept of being able to take something that for all intents and purposes can appear one way or another, and then turn that into how Chris closed the last lecture with, we are, we are seeing ourselves as both being entangled and also being separable. How did we go literally from a three dimension vertex to we're both entangled and we're, we're not living in a paradox. That's just the way it is. We're also removing ourselves or separating ourselves at the same time. That's what I would like to know because somewhere between three arbitrary lines and complexity is not a paradox. Something has to be happening and I'm not exactly sure how that transition plays out. Anybody want to enlighten me? Can you repeat your question please? Well, how do we, how do we go like from the, at this point of the reference frame description of saying it can't, you can't copy it, all you can do is pass it. Because we think things could be flipped based on internal internal reference frames, internal meanings and internal models. The thing we choose for context, yada yada yada, goes all the way out to the conclusion of Chris's conversation with us in number three, where he said, bottom line is, is that it's not a paradox that as things become more and more complex, then it wants to be able, it, it, it wants to get to a place of entanglement, alignment, a sense of agreement, and at the same time wants to remain separate. So between those two things, a whole lot has happened. But I don't think we've skipped over it. But we certainly haven't opened it up and really looked at it closely. And I'm just curious, what's in between those two scales? I agree. I think we're, we're gonna cover that towards the end of the, of the talk. But I agree. I mean, I don't know how you see this related to QRS, or the stuff we just talked about. But I think it's

the big question in this, in this lecture, right? That was like, where was this tension between increasing complexity and, you know, wanting, wanting to predict what the environment is doing? One requires entanglement, and the other one requires separability, at least colloquially speaking. I think this tension is basically what the free energy principle comes down to, right, Daniel? What do you think? You have the free energy, and you have the $E - TS$? One is prediction error. I mean, not exactly $E - TS$, right? That would be thermodynamics. But you know what I mean? One is prediction error, and the other one is, what was it? I can't remember.

Daniel Schrapp

I'll definitely look forward to the technical unpacking that you'll get to soon. But I was reminded of the William Blake quotation, do what you will, this life's a fiction, and is made up of contradiction. So let's modify that. Project what you will, rotate what you will, space times a fiction, not that it's false, but that it's a projected fictionalization, that it doesn't have the ontological primacy that the topology of communication does, and is made up of contradiction. Yes, and it's made up of this

tension between

fitting and not overfitting, between aligning and the principle of unitarity, and differentiating and becoming through others we become ourselves, and that type of articulation, and recognizing that two complementary forces like convex and concave come into existence at once, like a zero to two jump is not an issue or a stumbling block, it's that we wouldn't have it any other way.

Couldn't have it any other way.

Daniel Schrapp

Yeah.

Daniel Schrapp

Sure, we'll get that towards the end. You can almost even call it a duality, it's not far-fetched. I mean, it's the sort of thing that I think most people will be happy to take for granted, such, you know, supersymmetry, like you have two kinds of particles, but the same amount of them, and, you know, dualities that are all over the place. So I think it's fairly, it's not unreasonable to just accept that there are, there is a tension between increasing complexity, which you want, you want AMB to be separable for, and accuracy prediction. Now, how this is related to emergent space and this KRF stuff, you know, I don't want to make very hard-fetch claims. We'll talk about emergent space later in the course, but for now, let's say a few more words about KRFs, and then we'll get to the, you know, complexity stuff towards the end. So, so, KRFs are almost building blocks, if you want, for information processing. Here, it's very useful to recall the definition that I feel. I think we, we have mentioned before of information as being differences that make a difference. That's probably the best definition of information I've heard, and I think, I think it does a lot of work. I mean, it's almost mathematical, but anyways, we need to unpack it a little more and see. So, for a difference to make a difference, what do you want out of KRF, right? And surprisingly, I mean, not surprisingly, but reassuringly, I think would be the word. You need a KRF to have sensory,

action, and a default state, right? This is sort of like where your Bayesian prior distribution is encoding. These three things will give you a building block for any information processing. And that, you know, ties together with all the work in the pre-energy literature and perception, action, loops, and whatnot. So, before we see mathematically how KRF is defined, I think it's more illustrative. And I think Chris went exactly on the right order when he presented this a couple weeks ago, seeing examples from nature. These are actual pieces of hardware that are effectively QRFs. And it's remarkable how they map to... Well, I'll skip ahead a little bit. But when you get to the to the mathematical definition of a QRF as a quantum system that, you know, you can specify with this category theory diagram, you know, you have incoming arrows into a node that then expands, right? And this roughly corresponds to measurement and preparation degrees of... Not degrees of freedom. The degrees of freedom would be the same, right? When you attach it to the boundary. But measurement and preparation stages, I guess, would be the better word. But anyways, let me go back to these examples, right? And Daniel, you're the biologist here, so correct me if I say anything wrong here. You have a cartoon picture of a bacteria, I think. And this is a very simple sort of thing, right? It navigates in the direction of food. And that's it, right? So it is a very simple rudimentary example of a QRF. You have the sensory stage here, I guess, the degrees of freedom. I guess the degrees of freedom...

So it's not obvious here what the boundary is, that's one thing. But it is a good enough picture to see the resemblance to here. We have the measurement, which in this case, you're trying to measure concentration of food. You have your node here,

C, which is your prior distribution for concentration of food. And finally, you have the output, the preparation stage, which is your...

you know, the rest of this chemical network that will control and steer this bacteria, right? So there's here, you want to call it the action degrees of freedom that will steer this bacteria in the direction of food. And likewise with cell division, this I understand even less, the network appears more complicated. But I think it's fair to say that at the end of the day, all these chemical networks that appear super complicated somehow have to be mapped to this

category theoretical diagram. That's all there is to it. You're going to have the sensory input and the action output. And you're going to have the default chemical concentration, which amounts to this node over here. You remember this from last lecture, right, Daniel?

Daniel Kahn Yeah. And the cone-co-cone we explored also in Livestream 17 on the Bayesian semantic information flow. But I'll kind of add a point about these biological systems in relationship to artificially specified systems and what that helps us understand about perception, cognition, action. So the cone-co-cone diagram has this bow tie architecture, or this kind of like two pyramids touching point to point, where there's a preparation and a measurement, which can be different dimensionalities or different scopes. Like it could be multi-input, multi-output, or multi-input, single output, vice versa, and so on. But they come through this eye of the needle with C prime here. Now contrast that kind of architecture represented here abstractly with what we see in the bacterial process and in the cell division process.

These have that kind of a bow tie perception, cognition, action architecture, or measurement, cognitive integration, preparation architecture. In contrast, we see, for example, an artificial neural network where the input and the output layers are usually lower dimensional, maybe an image coming in with several hundred pixels and a single output, like a classifier coming out the other end, but a dimensional explosion in those internal cognitive layers. But in biological systems, it's actually exactly the opposite. For example, many, many, many receptors on the cell surface, on the holographic boundary, the sensorium of the cell. But then there are these common secondary messengers that many pathways converge upon. So that reflects the kind of radically different architecture that a QRF embodies, as opposed to the approach taken by disembodied intelligence frameworks, which use dimensional explosion internally to try to categorize and classify and find edge cases versus this dimensional winnowing that is reflective of these embodied intelligences.

And if I may make a tangential comment here also, I think the good thing about these two examples is that they also

showcase the fact of how, you know, the deep implications of a scale-free biology, if you want to call it this way, we

tend to think of these sorts of processes, right, of such as here, only happening at the level of the nervous system in animals,

or as you said, neural networks, right? We tend to associate this with neural networks.

And perhaps evolution has steered us in a way that's endowed us with nervous systems to make these things, maybe these are higher up in the

in the, in the, in the layers. But really there is, you know, that's a bit of an arbitrary distinction that we make, you know,

the processes going on at the cellular level, even if they don't appear to have anything to do with the nervous system,

my understanding is that they are, there are, there are some generic properties that are the same and everything that has to do with

information processing, you know, it manifests maybe slightly different, uh, when it comes to explicit nerves, but at the end of the day, it's the same generic process going on. Would you, would you say that's right, Daniel?

The, the ontological claim is that it's the same generic process happening,

whereas the more instrumentalist claim is that we can model it with some shared abstractions.

Yeah. Okay. So we can, we can continue, right? Um, so, uh, from, you know, I, I don't have a whole lot to say, Daniel

said quite a bit about, at the end of the day, QRF is, is, you can visualize it as this diagram, and,

you know, you can think of it as, uh, if you are more familiar with the picture of a neural network and ways on a neural network and so on,

that's the way to think about it. Now let's continue with explicit embodiment, if you wish. Uh, so QRFs in, in the

formalism have to be attached to, uh, a boundary, boundary degrees supreme. And I, I should add another

remark here. I think, and personally to me also, like the, the hard part about, uh, this quantum pre-energy principle formalism, if you wish, is, is what you said at the beginning is the fact that it's topological, right? So, so very often it's not clear what the boundary degrees of freedom are, right? Because we tend to think of boundaries as, you know, geometrical and, and certainly the classical FEP treats them as geometrical. Uh, but that's not the case here in the quantum pre-energy, uh, formalism and where we have emergent space, as will be talked about in later sessions. But that's just like one, one quick remark to make. At the end of the day though, even if it's topological, we're going to have real degrees of freedom that, um, this QRF is attaching to, this cone-cocone diagram. And you're going to have measurement operators that yield, um, classical outcomes, um, plus and minus one. And, uh, and then you can think of this QRF, um, or more abstractly, this cone-cocone diagram as, as a neural network, if it was, uh, acting on those boundary qubits. And then back propagating, uh, some sort of action signal preparation. Um, okay. Then let's not forget that, um, Landauer's principle and so on, uh, requires that we are mindful of energy constraints on computation. And that gives us the following picture.

Uh, number one, we're going to have to use energy, uh, from the environment to fuel our computation, to fuel our observation. So that means that right away you cannot use the entirety of your boundary to gather information. You're, there's, it necessitates a, uh, an allocation of resources, right?

Um, here E is the observed environment and the blue portion is a thermodynamical sector where you extract free energy and, and, and dump heat in order to run this computational process. And then furthermore, you can have, uh, a memory sector where you write things on the boundary. Okay. And this takes up energy, also, right? Writing information on, onto the boundary by Landauer's principle will take up information.

And, uh, Chris called this the minimal interest in observer because it has a memory sector and why minimal here, uh, implies that we have at least a minimal interesting implies that we have more than one layer here. The process of writing things into the, into a memory sector requires us to have a clock so that we can label those memories, um, in time.

Okay. So we do have time as part of this formalism, just not space yet. Okay. The time has to be internal to the observer.

Just, just one piece on that, that, that Chris mentioned, um, as, and as you echoed, time here is an attribute of some cognitive systems in terms of their internal understanding so that they can sort memories and understand what came earlier or later. It's not something that's a priori defined externally, which is the case in the kind of space time first. Now, how do we fit communication into here?

We flip that we're talking about the topology of communication as ontologically primal. And now certain kinds of sophisticated agents, it's not like necessarily true of a rock or a metronome or a particle of a gas, but some kinds of cognitive systems are such that they are able to generate an internal clock concept. And that doesn't have to be the same thing as what's happening on the wall with Kronos. This can reflect a purely internal time, Kairos, that enables observers or interactors

that have different clock structures. And then Landauer's principle helps us arbitrage or connect the informational demands to the energetic or the metabolic demands. Dean?

Dean? So when I, when I first listened to the lecture, I thought when the, when the term or phrase trade-off was raised, I, I traditionally have thought of trade-off as kind of a binary.

If I'm picking this, I'm trading off against all the other things that I didn't pick. So one or, or the other, or others, plural. But when I saw, when I saw this, and especially when I heard it in the context of things like degrees of freedom, I began to wonder whether it is, because it could sit on either side of that boundary. It could be a priori, or it could be ad hoc or post hoc, a trade-off.

Which prior to this conversation, I just assumed that a trade-off was one or the other. But essentially, what this implies is it, it isn't even, it's still a kind of a degree of, of how much weight you put into it. And then it, you kind of cross a threshold and then you've made your pick. You've, you've completed your trade-off. Or did I, again, did I misinterpret what I was hearing if it's, if it's topological as opposed to this, the more conventional space-time perspective? Is this, can, can a trade-off be both ad hoc and post hoc, seemingly at the same time?

Very interesting question. Let us just let it float and continue on. Enter.

All right. Because maybe, because again, if it's, if it, I'll just say this, if it, if it can be, that, that would really move us out of the classical realm.

That would essentially say that anytime we've made a pick, we re we really haven't made a pick.

There's still a whole bunch of other things that are going to happen that can, that can change whatever that action is that we thought we carried out or enacted. If we're talking about in, in an internal intern intern intern internality state, which is what I think we want. I think we want to be able to change our minds.

Let me take that one way, Dean. The, the analytical equations of the action perception loop don't tell us in implementation, like if we're going to write it in code, whether we should use action first procedurally to update our observations

or whether we enable our observations to, to wag the dog and then select action at the end of the loop.

So is it a perception action loop or is it an action perception loop?

and those enable a kind of seesaw where sometimes we want what we learn to change what we do and other times we want what we do to change what we learn and having that kind of a fulcrum or a pivot and again to connect it to the vertex popping in and out the analytical formalisms inscribed on the screen of the FEP like the textbooks they're ambiguous whether we lead with action and have inference follow or whether we lead with inference and and have action follow or or lead with action have inference follow whichever one of the combinations there are um because we and then that that um trade-off or runoff is broken by specific embodied cognitive agents that that trade-off about the um the procedural primacy of perception or action can't be broken generically right

it certainly makes for a lot of waste heat if you're not careful

it'd be exhausting a lot anyway

okay so um so i i see i i don't know if we can um talk about

intent um much with this formalism but we can take a very descriptive uh approach to it

and um go down here a little more uh actually before i don't want to get up ahead of myself

uh what i mean by that is

i whether we can put intent or not in this hierarchy um i i don't know i don't know if it matters a whole lot but at least we can talk about it you know operationally and descriptively and and say just like when you have a clock uh

you know that the clock is just there right like it is it if you don't have the clock you don't have the memory and that's it right uh and likewise here if we don't have attention uh an attention control no compartment we just cannot have we cannot play with all these qrfs uh that are non-commutative at the same time so if there is no attention control system to begin with well we don't we

cannot talk about a system that has non-commutative qrfs meaning here one one presumably goes to memory

sectors the other one goes to environment sectors and so on uh so

so this is certainly a necessity of the fact that we have um different sectors of the boundary being allocated to different purposes such as environmental observation and memory and so on um within the environment it was it's worth remarking that here if you if you think deeply about first principles and what it means to observe an entity um well you still come down to these differences that make a difference sort of thing right which involves whenever you're looking at the environment to if you're going to identify a system you need sort of a background that tells you that puts the system in context right so that you can identify the system as something distinct i think this again is something that we have talked about in previous uh discussions where the easiest picture here is something like the gauge speed gauge in your car where p is the gauge itself it tells you the speed but it's meaningless without the reference system right outside of outside of the thing that stays invariant right the context the background that gauge moving means nothing to you okay so we come on come back to this idea of if we think of information as differences that make a difference in terms of first principles this environment sector that we saw earlier has to be further broken down into pointer and reference states again this is covered uh in depth in the in chris's papers uh i'm thinking particularly of uh free energy for generating quantum systems

states um states um states um states um states um states um states um states um states um states um states um

likewise whenever we have so whenever your writing is happening to into a memory you need you need to have that clock whenever you are deploying non-community gear reps you're going to need uh attention and attention control compartment and we just described this operationally um if you didn't have it then the agent would not be what it is it could not deploy those non-community uh so marching on

uh we'll just end here with the krf alignment and the free energy principle just a few more words

uh this is possibly the most interesting picture in this in this uh presentation and just we can

recall here uh the possibilities for krf alignment and how free energy comes into play in this in this um formalism so is that we call we call a the trivial interaction we call this a thermal interaction and then you have a bunch of possibilities right and depending on each possibility you're you're going to see things as uh notes or as in the paper says uh non-local hidden variables uh the last situation is uh what we call alignment when we have krfs of the same size perfectly aligned okay so i can't quite recall i think this this would be the situation where you have uh noiseless classical uh computation right you just have a and b just interacting you know and uh it also also this is the situation where correct me if i'm wrong daniel this is this is entanglement right asymptotically once all the qubits are aligned you you have entangled and now we can tie back maybe two gains earlier so these two these two slides conclude the discussion basically and then we can open it up um so given this possibilities for how krfs can be aligned here is deployed by amd can be aligned uh and how you know in each situation you're going to get noise or or whatever um then what does the free energy principle say in this quantum setting well uh uh minimizing variational free energy leads you to entanglement okay so if theorem here if you are in the picture of b here and the qubits are aligned uh a and b are going to be intact now there is a tension between this tendency towards alignment uh as the free energy principle uh will carry you and uh observe reality which is that we're not just a blob of entangled particles there is separability and systems uh are at tension between maintaining separability um and at the same time being correlated enough to the environment so as to be able to predict predict it and now we can maybe go into a more philosophical discussion on the free energy principle um i should also say in the last month i just like i talked earlier at your apps um lenny saskin whose recent paper that talks about your app it led me into a rabbit hole of looking into his recent work and he's talked a lot about complexity

uh i'm being very speculative here so i
if someone can call me out uh that would be good in the comments but
just to skim it through some of his papers i saw a
monograph in 2018 where he talks about complexity
and you know he puts forward the second law for complexity
in quantum systems and increased complexity and it is very reminiscent of what's going on here you have a
tension between
uh systems tending to entanglement
uh
and the other side of the free energy functional if you wish
where systems
increasing complexity
and that necessarily requires
that a and b are separate
so daniel do you have any comments at this point
yeah there's a lot to it i i like that you
showed the four possibilities uh in the image above and and said it was a one of the most interesting
figures i i agree and uh
uh
how to connect this
to the classical fep
and also how to
see the quantum fep
not as just uh a second story built on top of the classical fep
which would be kind of like a uh historical
chronological understanding
of the theoretical apparatus
but it's almost like the quantum fep was like
a basement we didn't know about
because it it gives a broader context and though it appeared
conceptually several years later in the timeline we understand
that doesn't give it a secondary
um status
the classical fep
to be kind of like
a special case
of this
um

actually more

generic

setting

there's just a lot i'm very yeah uncertain maybe

i can i can i i can wrap it up uh and then if dean or you want to say something um

we can continue with that but let me conclude and this is all i have to say for today

um

um

i i think i mean

i'm sure chris will

explain this more in incoming uh sessions and lectures but i think the main

key differences is in the emergence of of space

of space how it might emerge from something more fundamental maybe a tensor network or who knows

uh but that's that's the big

big difference right and and i suppose the quantum free energy principle

!

!

you start to wrap your head around and and i i foresee interest in in those uh discussions and

lectures that uh deal with the concept of emergent space because every time you try to and i've done

it i've done this myself i've been thinking about how you could apply these formalism to

better known pictures or compared it for instance with thermal equilibrium with relaxation with thermal

equilibrium and then you start thinking about okay what is the boundary and so on

it's far from obvious right because again you have to remind yourself that the definition is topological

so in that regard i think it might be worth um

talking about one of the questions that came up that i thought i thought was good

and and then i can call it a day on this end um

so this question and i think it was the first question asked since the last lecture

can you just read the full question and then read the full answer thank you

the question and the answer it's number 28 here can you guys see it should i zoom in a little bit

it's it's pretty small but but reading it will be good

okay so this question i think i've asked myself this question

too um maybe with different in a different way but basically you know it gets to the point of what i was

saying earlier right um that it would be nice to see how we can use this quantum free energy principle

formalism to recover maybe familiar settings such as relaxation to thermal equilibrium via the heat equation but that's far from always how you do it if your definition is topological right

so the question says what happens when the agent becomes identical to the environment

uh when the model has been improved so much that it does not differ from the environment is this a stadium impossible would it be through my amical equilibrium maximum entanglement okay so clearly it's not thermodynamical equilibrium but i think what the person is asking if i maybe if i'm understanding this correctly they're wondering okay so if the free energy principle drives a and b towards entanglement then is it fair to say that they have that any distinction between them has dissolved right that they have merged into the same thing um

and that has to do with this whole definition of philosophical debate of thingness that uh carl priston you know is a proponent of right uh what what it means for two things to be separate okay so let's see what chris said um

uh christian has pointed out that the limit of the classical fp the limit of perfect prediction corresponds to generalizing currently between system and environment each sends messages that the other can perfectly predict uh now however it's clear and i'm not reading anymore i'm i'm making a quick remark here i'll continue in a second in the classical fp it's clear that a and b even if they are you know in the perfect limit of the fp they remain different the different entities right because you have that if you wish that extra

uh layers of differentiation coming from their spatial embedding right in the in the classical fp that is clear that they haven't even if they are getting synchronized they have not lost their separate identities you can always say well this is the one that was to the left and this is the one that was to the right pun intended with earlier discussion right um you have that extra uh resource for for separating them if you wish by virtue of the spatial embedding right the spatial embedding is giving you this extra information that allows you to keep track of of of of a and b separately okay so i'll continue this notion depends on the system and its environment being separated and has distinguished somehow in classical formulations they are separated in ordinary 3d space i.e they occupy different locations so they are different because they occupy different locations right but if you don't have that geometrical background then then how you know that they're different right okay so i continue the quantum formulation is background free meaning it assumes no space-time embedding system and environment

do not have different locations even though they have different sets of degrees of freedom in the overall hilbert space here the fvp the limit of the fvp is maximum entanglement this does not mean that the system and environment are identical they are and since they have different degrees of freedom okay so this is just addresses the question i i just posed it means that the joint state is not separable okay um into a system s and environment t their states are no longer conditionally independent this is distinct from thermal equilibrium which means that they have the same temperature while remaining separable and that their interaction can be characterized as the change of thermal fluctuations i.e noise so i i guess this is a nice warm-up for things that are to come but i i do think that this whole

uh idea of emergent space is going to be um possibly you know one of one of the most important in the course

and it's going to help stick out between the classic and quantum formulations for sure

dean what's your take on this what what i what i saw what i saw was that when you have that kind of alignment that perfect alignment that is one that was one that was d in that diet in that

that diagram right that was one state of a of a complex state that included an a and a b and a c and probably a whole bunch of other ways that i mean yeah right so that one state where you've got this kind of entanglement isn't isn't into perpetuity like you don't get there and then stay there and that's where i think the question of separability and complexity

it still works because just because you momentarily gotten to a place that we define in here as d doesn't mean you stay there indefinitely you could be swapping around to b and c and a in the next instant so that's that at least that holds the complexity piece up that that moment for however long that moment is where x a and x b are aligned isn't permanent

so it's it's not predictable in that sense therefore it is complex i still don't know how we get though that the closer we get to agreement the more complex things are

unless we're just talking about d in this array of a b c and d

that goes back to that quantum reference frame i think like i said i again maybe i'm interpreting this wrong but when i message pass the reason why i have to like literally give it over to the other person and let them have it i can't give have them copy what i'm doing is because there has to be i guess that reinforces the separability piece of this as well the fact that i can't give them a copy that i can only give them the information and then they've got to sort of unpack it in their own in their own way of taking it off the the boundary but again i struggle with this i'm i'm guilty as charged but i am i am what i'm trying to do is trying to come up with my own narrative so that what chris is saying makes sense to me as opposed to make sense to the world which is what chris is trying to do i'll try with another speculative fiction so um there's a screen between let's just say the uh stage and the audience or whatever it's a holographic screen and on each side of the screen there's some embodied object so that means that it must have um vault it's a volume-enclosing object because points and lines and planes are only abstractions every real physical object encloses volume and now a light is going to be coming from behind each object facing the screen and that's this kind of uh cone that we're seeing from each side it's like the shadow cast by the object so if the object was a sphere then it would be projecting down to a circle onto the screen if the shape were a cube it could be projecting down and so where the light is positioned is the um perspective that that cognitive entity is taking on its qrf those lights could move so now we have this situation where the two spotlights are moving maybe the shapes are staying the same maybe they're changing too and it's like two search lights scanning across this b oval and some sometimes they are perfectly locked i mean there could be a situation where they're perfectly aligned or there could be situations where they're totally disjoint um

and each of those situations have different implications for what one entity sees when it looks at the screen maybe like the shadow is the only part where you can see because the rest of it is too blinding or something

like that

and so then if we're in case a here where um maybe b doesn't have a projection or maybe b's projections totally somewhere else then a is in this solipsistic i'm the active agent interacting within an inert world setting and d is like potentially the um the computer engineer's classical dream or asymptote which is like all the data going through the usb cable is just being perfectly noiselessly received and then we also get these um b and c cases but a is kind of like the inner world d is like the perfect communication channel and then to connect it just to Fristen's concept of generalized synchrony as we've talked about many times generalized synchrony doesn't mean lockstep two agents could reduce

their uncertainty about each other not by doing the same thing at every time step but for example by turn taking

and i think that points the way to how we get reduction of variational free energy reduction of prediction error surprise and so on without especially spatial temporal enmeshment but not necessarily even informational identity in measurement because you could have two things that through their alignment become actually differentiated and that could increase the mutual information between two metronomes not just to have them going in the same direction but to have them going in opposite directions the mutual information those two settings is going to be high so it's not the case that generalized synchrony means that um things are becoming uh blurred we could actually also have integrated information through increased articulation not just decreased and then life finds a way or cognition finds a way on this trade-off frontier of in our pursuit of unitarity will we blur to unify or will we differentiate to unify but those are the paths we have we don't have the um backwards ratchet

yeah yeah makes sense

any other ladder slides enter uh no that's it uh so chris made a few more remarks at the end about uh language uh language here uh but he mentioned that he's gonna talk more about this in the coming lecture so we'll just leave it out to him

yeah there could be some very powerful and interesting linguistic opportunities for updating our speaking to reflect what we are learning here

um

i don't know what those necessarily are

whether it's a syntactic or a semantic thing or whether it's a narrative thing but how to actually re-enter some of these formalisms into our

our focus understanding could just be immensely powerful and um we could have a lot of integrity between the natural language way that potentially were already showing up for engagement and some of these formalisms that get to first principles about interacting entities and boundary conditions and all of that dean any penultimate thoughts

not really i i'm going to go back and listen to the

i'm going to listen to chris's comments again about this common cause

and the fine-tuning assumption and whether that's sort of a metaphorically a dialing in or what it

what he what he means exactly by the fine tuning in the in the context of this conversation today

what you just kind of said about are we dulling something here or are we sculpting something to make it more fittable like i'm going to go and now and go back and listen to what he talks about in terms of fine tuning and maybe

update my priors around that so

and are any last uh thoughts or maybe scroll over to the um schedule of the course again just so we can close on that

yeah we're halfway through we just completed the third discussion section now so we've had three lectures and three

discussions it's all downhill from here um by the way this was the first midterm

you guys both did well um we're gonna head into four or like session four with these last three how

agents use multiple communication channels and that's going to get connected to the quantum fvp

how space time emerges from communication which is something that we've brought up a bunch it's been in the background but but making that explicit is going to be exciting and then section six

and then section six i think is just going to be such a jumping off point because the scope

and the intensity of what is transpiring is extreme so connecting it to biology

which is to say life more fully and where we go in in 23 and beyond that's going to be awesome so any last words enter

uh no i just think of what you said um very i would i would say it's going to be uphill in in terms of how interesting it's going to get

i feel in a good sense meaning that we're planning cool well i encourage people to uh check out the course website and submit questions that they'd like to see addressed by chris also to register and join

these participatory discussions there'll be three more of them so if you want to join share your qrf

be part of the publication of all these works that will compile at the end and just get your exoskeleton

in the game with what is happening here it'd be really awesome to hear more voices and and to have

more on board okay thank you dean and andre till next time

is